Pixie: AI-Powered Photo Gallery Supplementary Document

Submissions:

Project Demo: https://www.youtube.com/watch?v=5mEDDtHzC_w

Supplementary Document: Kiddocare

Github Repository: https://github.com/RahulRR-10/Kiddocare_RVCE_4_B.AmishaPai.git

Presentation Slides:
https://www.canva.com/design/DAGyphpqxc8/eQnkCYDO6u_apMiAT-2wnQ/edit?utm_content=DAGyphpqxc8&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

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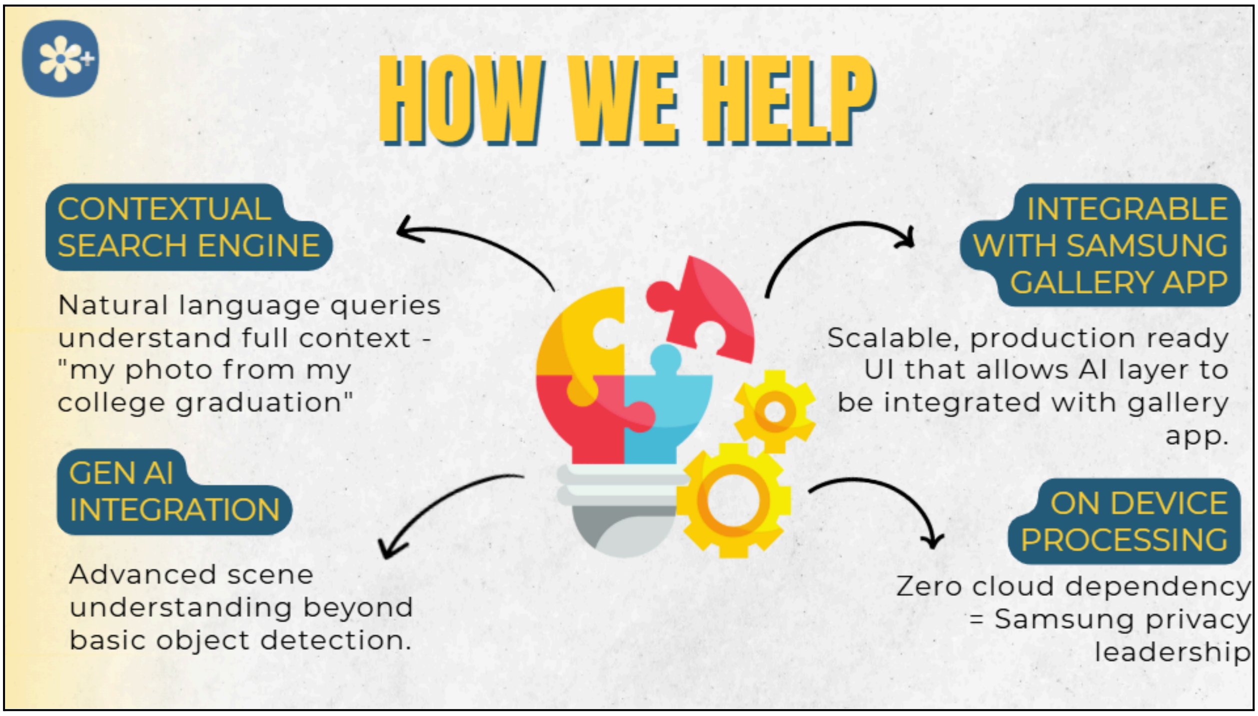
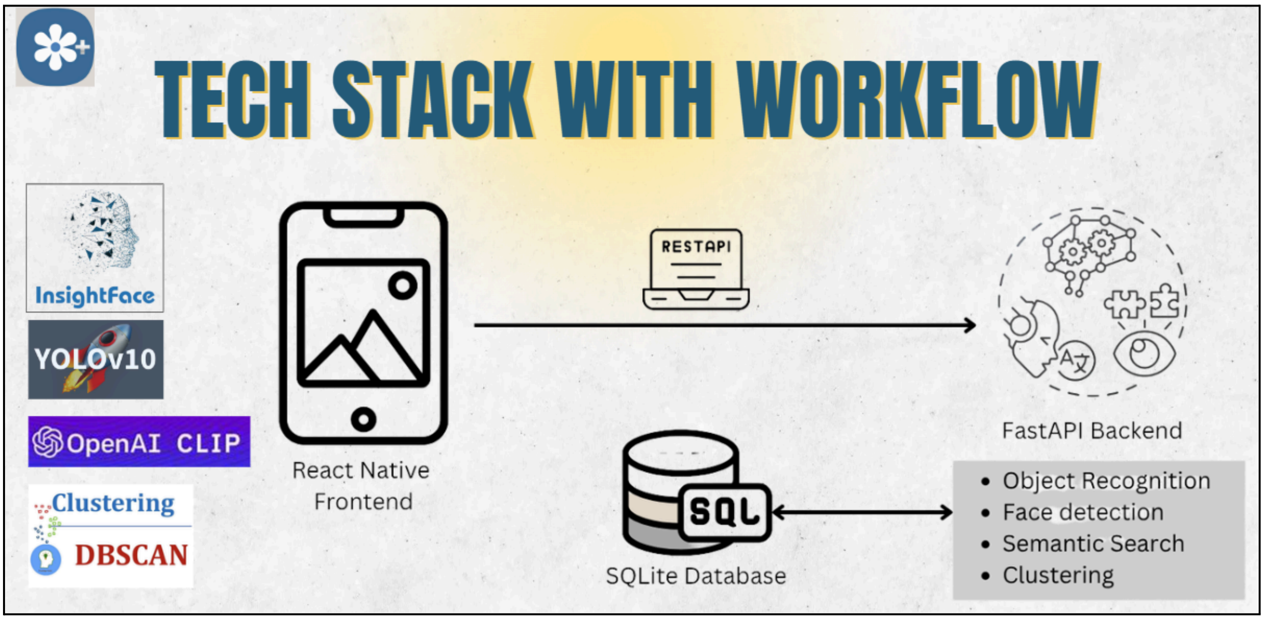
Why Now? With 1.4 trillion photos taken annually and 73% of users preferring local processing for privacy, there's a massive opportunity for an on-device AI solution that combines convenience with privacy. Samsung's hardware capabilities make this technically feasible for the first time.





Stakeholder Impact

For Consumers:

- Effortless photo discovery through natural language
- Complete privacy preservation with on-device processing
- Superior search accuracy compared to cloud-based competitors

For Samsung:

● Differentiated privacy-first AI capabilities
● Reduced cloud infrastructure costs
● Enhanced Galaxy device value proposition against iPhone and Pixel

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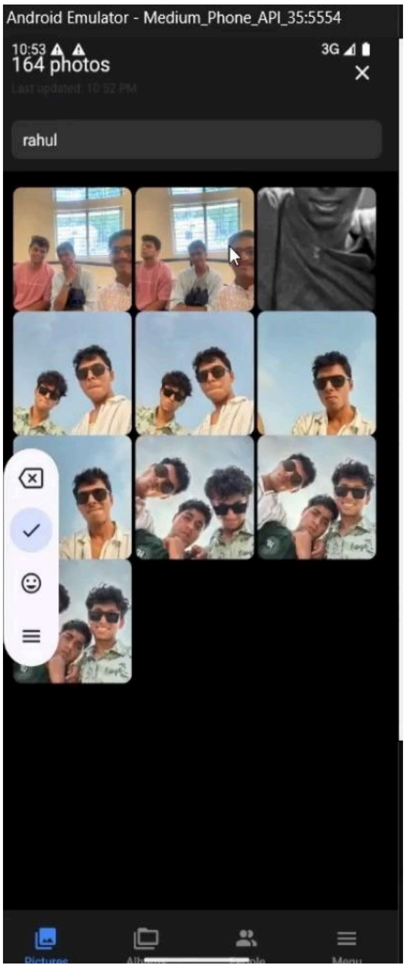
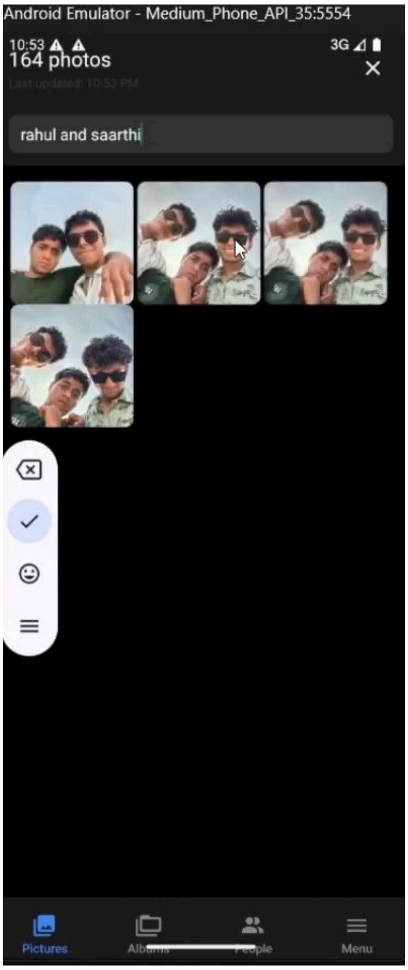
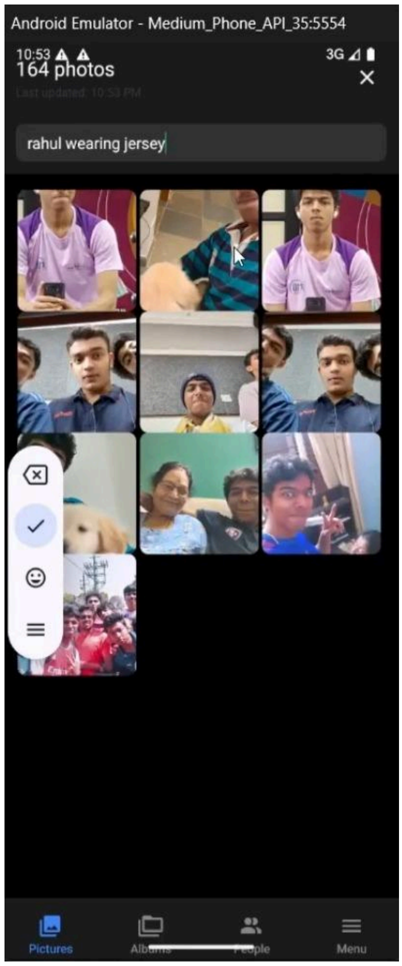
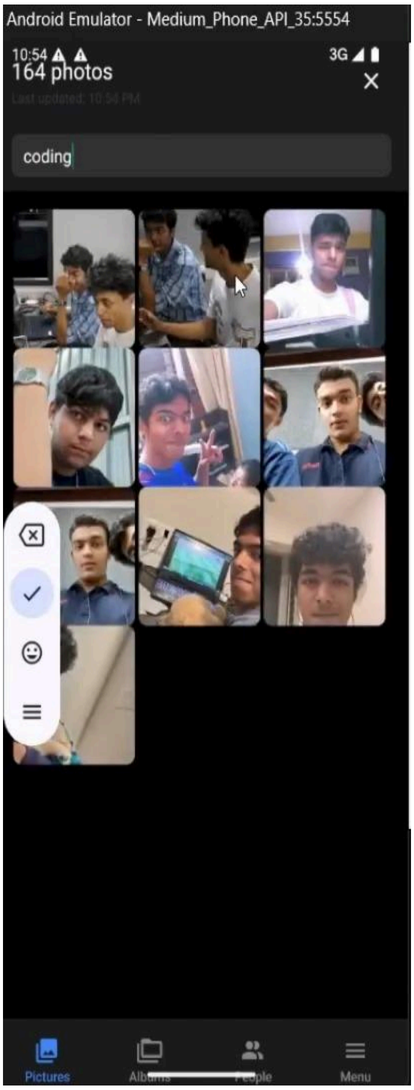
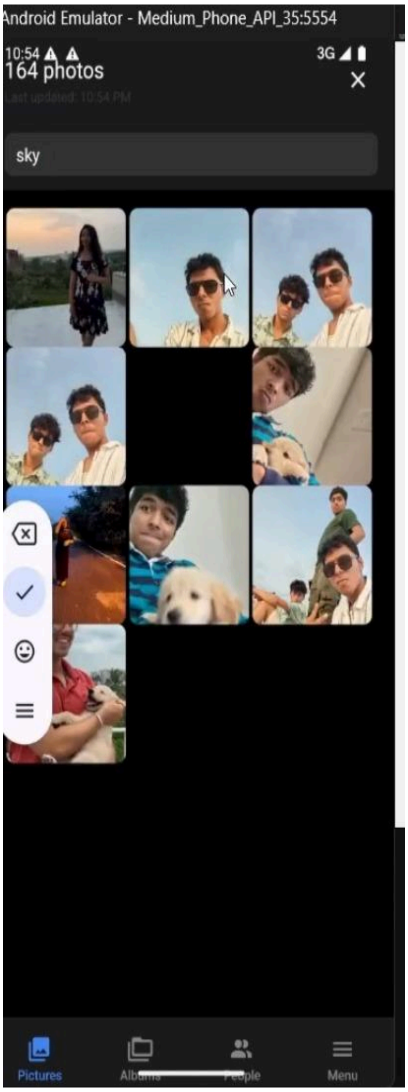
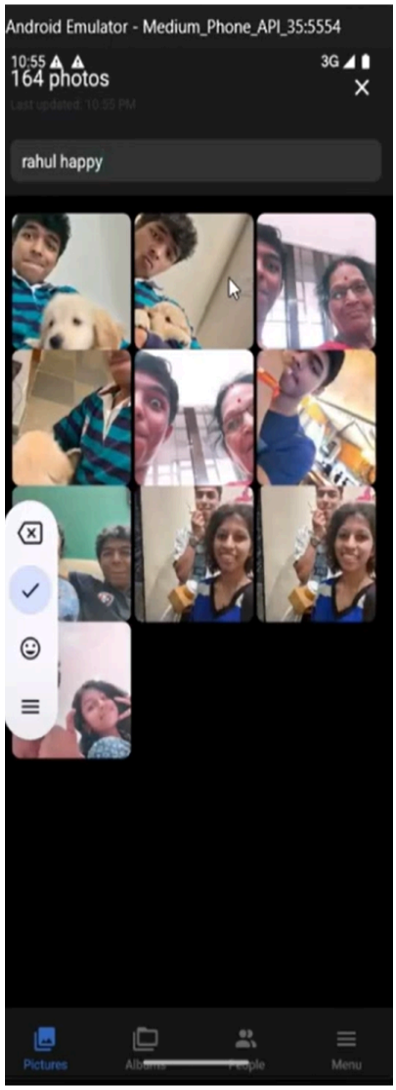
Domain Mastery: End-to-End Search Excellence

Query Versatility: Handling 10+ distinct query types from simple to complex

Table demonstrating- Query, Feature, and the Screenshot of Proof

“Rahul”	"Rahul and Saarthi"	"Rahul wearing jersey"
Direct facial embedding matching	Group Intelligence & Relationship Discovery	Person + Clothing Detection
		

<table>
<tr><td>"coding"</td><td>"Sky"</td><td>"Rahul happy"</td></tr>
<tr><td>Activity Detection</td><td>Object & Scene Recognition</td><td>Emotional State Detection</td></tr>
<tr><td>
</td><td>
</td><td>
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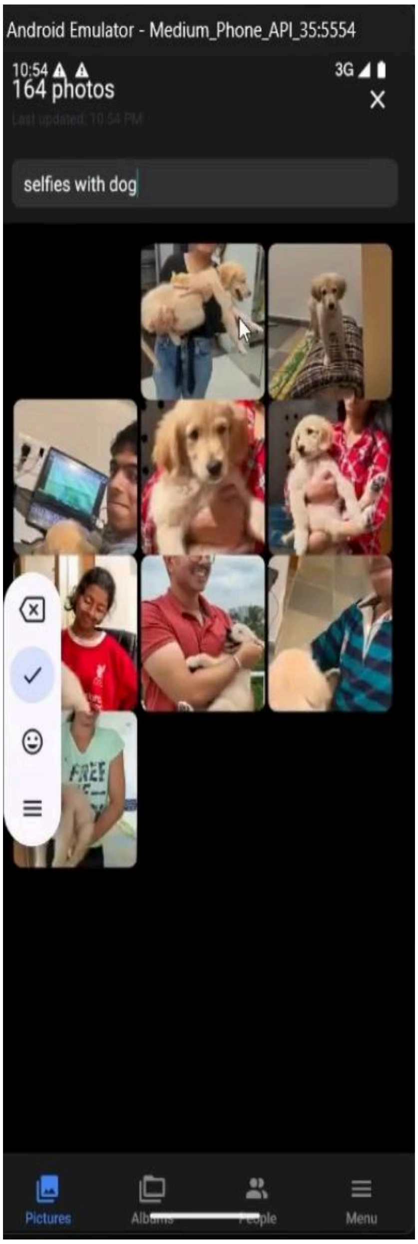
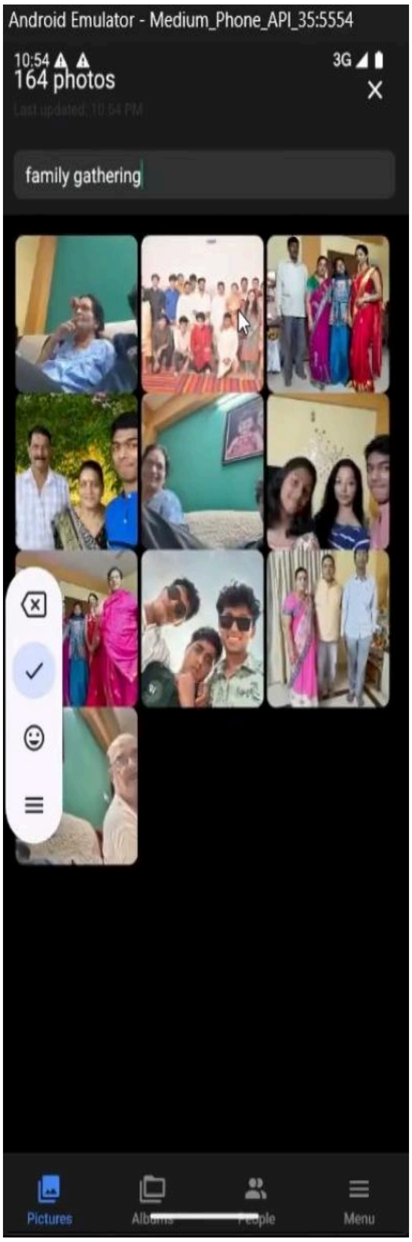
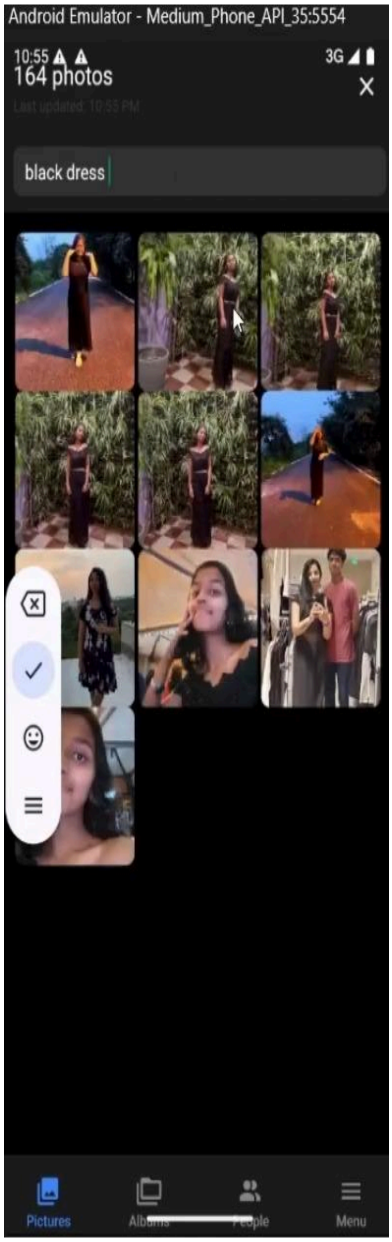
<tr><td>"Selfies with dog"</td><td>"Family Gathering"</td><td>"Black dress"</td></tr>
<tr><td>Photographic Concept Recognition</td><td>Contextual Scenario Intelligence</td><td>Color-Based Queries</td></tr>
<tr><td>
</td><td>
</td><td>
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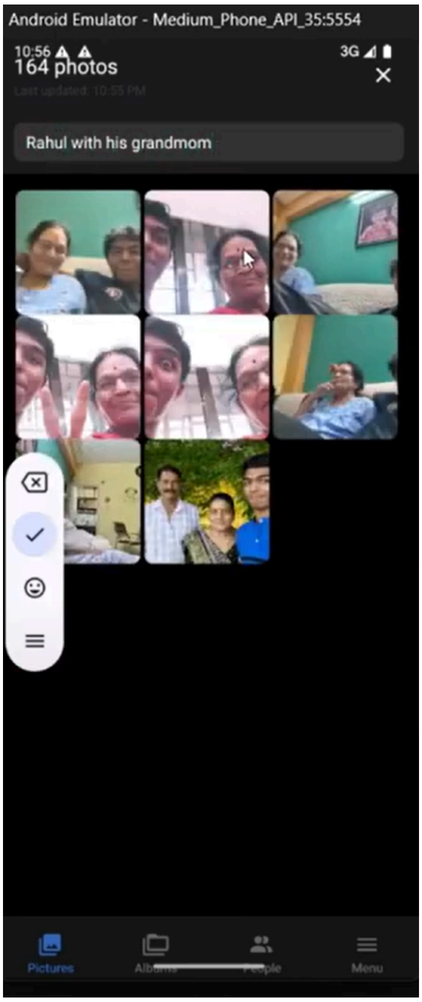
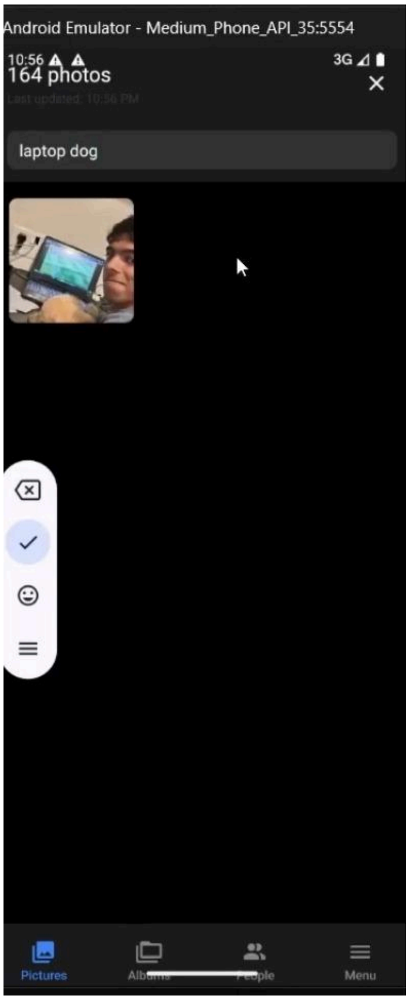
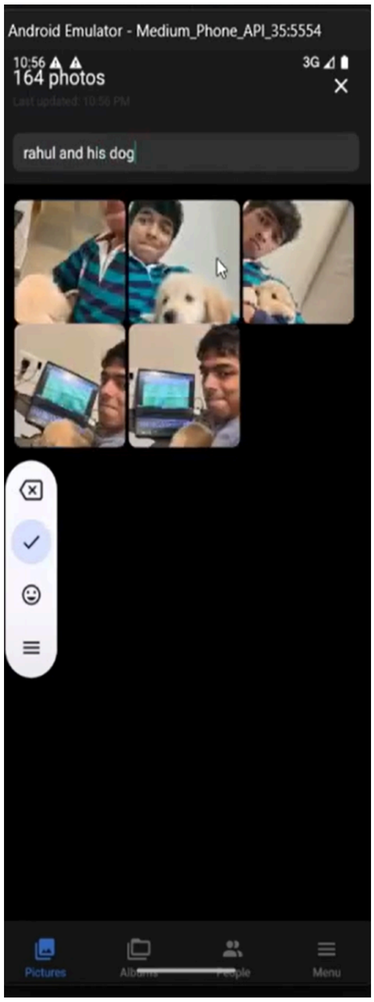
"Rahul and his grandmom"	"laptop and dog"	"Rahul and dog"
Familial Relationship Recognition	Multi-Object Detection	Multi-Entity Detection
		

CHALLENGES & SOLUTIONS

Challenge	Problem	Solution
Multi-Modal Query Processing Complexity	Creating a unified search interface that could intelligently route natural language queries to appropriate AI models while combining results coherently. For instance, "*Rahul wearing a jersey*" requires person recognition (InsightFace) + clothing detection (YOLOv10) + semantic understanding (CLIP) working in perfect harmony.	Developed query analysis engine that parses natural language inputs and dynamically determines which models to invoke. Technical Implementation: • Query preprocessing to extract entities (people, objects, contexts) • Dynamic model routing based on query complexity • Confidence-weighted result fusion using cosine similarity • Caching layer for frequently accessed embeddings

Breakthrough: The system uses confidence scoring and weighted fusion algorithms to combine results from multiple models. It implements a contextual understanding layer that recognizes query patterns and optimizes model selection accordingly.

FUTURE PLANS

01 Advanced Temporal Intelligence

Enabling queries like "how has Sarah's appearance changed over the years?" This would involve temporal embedding spaces and change detection algorithms.

02 Cross-Device Synchronization

Develop a privacy-preserving synchronization system that allows multiple Samsung devices to share AI-processed metadata without compromising the core on-device processing principle. This could use homomorphic encryption for secure embedding sharing.

03 Integration with Bixby AI

Create an AI system that proactively surfaces relevant memories based on context - current location, calendar events, or contact interactions. For example, automatically showing beach photos when planning a vacation or family photos before a gathering.